Supplementary Material

Spectral Co-Design of Agrivoltaic Modules for Multifunctional Net Energy Benefit: A Quantum-Guided Digital Twin

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S1 8-Site FMO Hamiltonian and full coupling matrix

S1.1 Hamiltonian formulation

The 8-site Fenna–Matthews–Olson (FMO) Hamiltonian used throughout this work extends the standard 7-site model [1] by including the eighth bacteriochlorophyll *a* molecule that bridges the chlorosome baseplate to the reaction center [2]. The full electronic Hamiltonian in the site basis is:

$$\hat{H}_{\mathrm{FMO}} = \sum_{n=1}^{8} \varepsilon_n \, |n\rangle\langle n| + \sum_{n \neq m} J_{nm} \, |n\rangle\langle m| - i\Gamma_{\mathrm{RC}} \sum_{j \in \{3,4\}} |j\rangle\langle j| \,, \tag{S1}$$

where ε_n are the site energies (referenced to the lowest-energy site), J_{nm} are the inter-site electronic couplings, and $\Gamma_{\mathrm{RC}} = 0.15\,\mathrm{fs}^{-1}$ is the effective irreversible reaction-center trapping rate applied to Sites 3 and 4. The anti-Hermitian term $-i\Gamma_{\mathrm{RC}}$ models irreversible charge separation at the reaction center; the Hamiltonian is non-Hermitian only in these diagonal trapping terms.

S1.2 Site energies

Table S1 lists the site energies used in all production simulations. Energies are given in cm^{-1}, both absolute and relative to the lowest-energy site (Site 3 at $12\,210\,\mathrm{cm}^{-1}$).

Table S1: **8-site FMO site energies** (in cm^{-1}, referenced to Site 3).

Parameter	**Site 1**	**Site 2**	**Site 3**	**Site 4**	**Site 5**	**Site 6**	**Site 7**	**Site 8**
Absolute ε_n	12 490	12 630	12 210	12 320	12 480	12 710	12 520	12 410
Relative ε_n	280	420	0	110	270	500	310	200

The relative site energies follow the Adolphs–Renger parameterization [1] for Sites 1–7; the absolute scale is fixed by setting the lowest-energy site (Site 3) at $12\,210\,\mathrm{cm}^{-1}$. Site 8, which extends the antenna manifold above the trapping pair, lies $200\,\mathrm{cm}^{-1}$ above Site 3 ($\varepsilon_8 = 12\,410\,\mathrm{cm}^{-1}$ absolute) based on the eighth-bacteriochlorophyll characterization by Moix *et al.* [2].

S1.3 Electronic coupling matrix

Table S2 presents the full 8×8 electronic coupling matrix J_{nm} (in cm^{-1}). The matrix is symmetric ($J_{nm} = J_{mn}$), and all 28 unique coupling pairs are listed. Values for Sites 1–7 follow the Adolphs–Renger parameterization [1]; couplings involving Site 8 are taken from Ref. [2]. This site-8 block is held fixed across all production runs: its strongest link, $J_{18} = 21.0\,\mathrm{cm}^{-1}$, attaches site 8 to the antenna, while its link into the trapping pair, $J_{38} = 0.6\,\mathrm{cm}^{-1}$, is the weakest entry in the matrix — two orders of magnitude below the J_{12} backbone ($|J_{12}| = 87.7\,\mathrm{cm}^{-1}$; entry-channel link $J_{18} = 21.0\,\mathrm{cm}^{-1}$ is about one quarter of J_{12}). This weakness is why the 7-site filtered- and broadband-excitation references carry over unchanged to the 8-site network (main text).

S1.4 Reaction center trapping

The irreversible trapping is implemented through a non-Hermitian term on Sites 3 and 4, which together form the primary funnel to the reaction center in the 8-site FMO model [3, 2]. The trapping rate $\Gamma_{\mathrm{RC}} = 0.15\,\mathrm{fs}^{-1}$ is the effective rate used in all production runs; it corresponds to a capture time $\tau_{\mathrm{trap}} \approx 6.7\,\mathrm{fs}$, short enough that near-complete passive capture is physically plausible within the 1-ps window of Eq. (S2); the quantitative reference used below ($\Phi_{\mathrm{FT}} = 0.98$) is the assumed passive capture-efficiency cap — a model input, not a trajectory integral (see the trace-conservation paragraph of Section S11).

Table S2: **Full 8-site electronic coupling matrix** J_{nm} (in cm^{-1}). The site-8 column (strongest link $J_{18} = 21.0\,\mathrm{cm}^{-1}$ into the antenna, weakest link $J_{38} = 0.6\,\mathrm{cm}^{-1}$ into the trapping pair) extends the antenna about four times below the J_{12} backbone (vs. $|J_{12}| = 87.7\,\mathrm{cm}^{-1}$), so site 8 modifies the entry channel without opening a fast decay path — the regime held fixed across all production runs.

	Site 1	**Site 2**	**Site 3**	**Site 4**	**Site 5**	**Site 6**	**Site 7**	**Site 8**
Site 1	0	−87.7	5.5	−5.9	6.7	−13.7	−9.9	21.0
Site 2		0	30.0	8.2	0.7	11.8	4.3	−4.2
Site 3			0	−53.5	−2.2	−9.6	6.0	0.6
Site 4				0	−70.7	−17.0	−63.3	−1.3
Site 5					0	81.1	−1.3	1.5
Site 6						0	39.7	−7.9
Site 7							0	12.0
Site 8								0

The forward transfer yield is defined as:

$$\Phi_{\mathrm{FT}} = \Gamma_{\mathrm{RC}} \int_0^{t_{\max}} [P_3(t) + P_4(t)]\, dt, \tag{S2}$$

with $t_{\max} = 1\,\mathrm{ps}$ and $P_j(t) = \rho_{jj}(t)$. Both trapping sites enter the integrand.

S1.5 Hermiticity validation

The non-Hermiticity is strictly confined to the diagonal entries of Sites 3 and 4. For all $i \neq j$, $H_{ij} = H_{ji}^*$ holds within machine precision (1×10^{-14}). We verify the Hermiticity condition for all 64 matrix elements at every time step of the simulation.

S2 Convergence analysis

S2.1 Hierarchy depth convergence

We adopt $L = 8$ as the production standard to ensure robust convergence across all disorder realizations. The PT-HOPS/SBD hierarchy implementation is inherited from the validation framework of Ref. [4], where convergence was established by MAE analysis on the same excitonic parameters. The convergence audits (hierarchy-depth and Matsubara MAE values, and the 2.3 % detailed-balance deviation) are inherited from the 7-site companion study [4] and are re-quoted here; they characterize the solver protocol rather than the 8-site extension.

Table S3 summarizes the mean absolute error (MAE) between successive L values and the relative deviation from the $L = 9$ reference (computed for a subset of 10 disorder realizations).

Table S3: **Hierarchy depth convergence.** MAE between successive hierarchy depths for exciton populations, averaged over all 8 sites and the companion study's audit ensemble ($n = 10$).

L	MAE ($L \to L+1$)	Rel. dev. from $L = 9$
6	9.8×10^{-10}	1.2×10^{-8}
7	9.6×10^{-10}	1.1×10^{-8}
8	3.1×10^{-11}	4.2×10^{-10}

The MAE between $L = 7$ and $L = 8$ is 3.1×10^{-11}, which is negligible at the precision of all reported observables.

S2.2 Matsubara truncation

We test Matsubara term counts $K = 1, 2, 3$ to assess the impact of high-temperature corrections. Table S4 shows the effect on the coherence lifetime τ_c and the forward transfer yield Φ_{FT}. Here τ_c is the $1/e$ decay time of the l_1-norm coherence (defined in the main text), and $\pm$ denotes one standard deviation over the trajectory/disorder ensemble (n as stated).

Table S4: **Matsubara truncation convergence.** Impact of K on key observables at 295 K under dual-band filtered excitation; the broadband reference under identical conditions is $\Phi_{\mathrm{FT}} = 0.71(2)$ (main text and Ref. [4]). Ensemble size $n = 100$ (parenthetical values are 1 standard deviation).

K	τ_c (fs)	Φ_{FT}	MAE from $K = 3$
1	406(30)	0.86(3)	3.4×10^{-4}
2	420(35)	0.89(3)	3.3×10^{-5}
3	422(36)	0.89(3)	—

The MAE between $K = 2$ and $K = 3$ is 3.3×10^{-5}, confirming that $K = 2$ is sufficient for converged dynamics at room temperature. At 295 K, the dynamics are dominated by the explicit vibronic modes rather than by high-order Matsubara corrections to the Drude–Lorentz background, as expected.

S2.3 Time step convergence

We verify time-step convergence by comparing $\Delta t = 0.2$ fs, 1.0 fs and 2.0 fs. Table S5 summarizes the results.

Table S5: **Time step convergence.** Maximum population deviation relative to the $\Delta t = 0.2$ fs reference.

Δt (fs)	Max. population deviation
0.2	— (reference)
1.0	4.3×10^{-3}
2.0	8.1×10^{-3}

We adopt $\Delta t = 0.2$ fs as the production standard, consistent with Ref. [4].

S2.4 Detailed balance

We compare the steady-state populations with the analytical Boltzmann distribution to verify detailed balance. Because the production Hamiltonian carries an irreversible trap, this audit is run on the trap-free reference and is inherited from the companion propagator (Section S2); it characterizes solver thermalization rather than the trapping model itself. The maximum deviation across all 8 sites is 2.3 %.

S2.5 Physical consistency checks

All production simulations satisfy the following consistency checks:

- **Trace preservation,** $\max_t |\,\mathrm{Tr}[\rho(t)] - 1| < 1 \times 10^{-12}$
- **Positivity,** $\min_i \lambda_i(t) > -1 \times 10^{-14}$ for all eigenvalues λ_i of $\rho(t)$
- **Hermiticity,** $\max_{i,j,t} |\rho_{ij}(t) - \rho^*_{ji}(t)| < 1 \times 10^{-14}$

- **Population realness,** $\max_{i,t} |\mathrm{Im}[\rho_{ii}(t)]| < 1 \times 10^{-14}$

S3 Polaron dressing mechanism and residual reorganization energy

S3.1 Canonical (Lang–Firsov) transformation

The polaron dressing mechanism that underlies the coherence protection reported in the main text can be rigorously understood through the canonical Lang–Firsov transformation [5, 6]. The full system-bath Hamiltonian is:

$$\hat{H} = \hat{H}_S + \sum_n |n\rangle\langle n| \sum_k g_{n,k}(\hat{b}^\dagger_{n,k} + \hat{b}_{n,k}) + \hat{H}_{\mathrm{bath}}, \tag{S3}$$

where $g_{n,k}$ are the system-bath coupling constants for mode k at site n, and $\hat{b}_{n,k}$ ($\hat{b}^\dagger_{n,k}$) are the bath annihilation (creation) operators.

The unitary transformation $\hat{U} = \exp\left[\sum_n |n\rangle\langle n| \sum_k \frac{g_{n,k}}{\omega_k}(\hat{b}^\dagger_{n,k} - \hat{b}_{n,k})\right]$ partially diagonalizes the system-bath coupling. After applying $\hat{U}$, the transformed Hamiltonian becomes:

$$\hat{H}' = \hat{U}^\dagger \hat{H} \hat{U} = \hat{H}'_S + \hat{H}_{\mathrm{bath}} + \hat{H}_{\mathrm{res}}, \tag{S4}$$

where $\hat{H}'_S$ contains renormalized excitonic couplings (the polaron-shifted energies) and $\hat{H}_{\mathrm{res}}$ represents the residual system-bath coupling after the transformation.

S3.2 Residual spectral density

Underdamped vibronic modes that satisfy the resonance condition $\Delta E_{nm} \approx \hbar\omega_v \pm J_{nm}$ become partially "dressed" by the canonical transformation: they are incorporated into $\hat{H}'_S$ and no longer contribute to dephasing in the transformed frame. The residual spectral density is:

$$J_{\mathrm{res}}(\omega) = J_{\mathrm{bath}}(\omega) - \sum_{k\in\mathcal{S}} \frac{\lambda_k \omega_k^2 \gamma_k}{(\omega - \omega_k)^2 + \gamma_k^2}, \tag{S5}$$

where $\mathcal{S}$ denotes the set of underdamped modes that are vibronically dressed.

S3.3 Quantitative effect

For the 12-mode Kleinekathöfer/Coker parameterization at 295 K ($\lambda_{\mathrm{DL}} = 35\,\mathrm{cm}^{-1}$ and $\sum_{k=1}^{12} S_k\omega_k = 92.6\,\mathrm{cm}^{-1}$), the residual reorganization energy under the two excitation scenarios is:

$$\lambda_{\mathrm{res}}^{\mathrm{broad}} = \lambda_{\mathrm{DL}} + \sum_{k=1}^{12} S_k\omega_k = 127.6\,\mathrm{cm}^{-1}, \tag{S6}$$

$$\lambda_{\mathrm{res}}^{\mathrm{filt}} = \lambda_{\mathrm{DL}} + \sum_{k\notin\mathcal{S}} S_k\omega_k = 110.2\,\mathrm{cm}^{-1}, \tag{S7}$$

where the dressed set $\mathcal{S}$ contains the manifold members bridging the trapping sites to the bright exciton. The site energies fix these bridges arithmetically: $12\,500\,\mathrm{cm}^{-1} - 12\,320\,\mathrm{cm}^{-1} = 180\,\mathrm{cm}^{-1}$ (site 4) and $12\,500\,\mathrm{cm}^{-1} - 12\,210\,\mathrm{cm}^{-1} \approx 290\,\mathrm{cm}^{-1}$ (site 3, matched by the $280\,\mathrm{cm}^{-1}$ member), so $\mathcal{S} = \{180, 280\}$ and $\sum_{k\in\mathcal{S}} S_k\omega_k = 17.4\,\mathrm{cm}^{-1}$. The red 820 nm passband ($100\,\mathrm{cm}^{-1}$ full width) is centred on site 3 (819 nm) and lies within $75\,\mathrm{cm}^{-1}$ of site 4 (812 nm), while the blue 750 nm passband lies $833\,\mathrm{cm}^{-1}$ above the $12\,500\,\mathrm{cm}^{-1}$ centre — between the $720\,\mathrm{cm}^{-1}$ and $1\,050\,\mathrm{cm}^{-1}$ members, and hence outside every manifold resonance — acting as a transparent control window.

This matches the companion band scan, in which moving the blue edge from 730 nm to 770 nm left Φ_{FT} unchanged whereas removing the red band collapsed it. Because pure dephasing proceeds as $\tau_c \propto \lambda_{\mathrm{res}}^{-2}$ in the high-temperature limit, the 13.6 % reduction of λ_{res} predicts a coherence extension of $\times 1.34$, against the computed $\times 1.50(18)$ (280 fs $\rightarrow$ 420 fs) — consistent within the combined 1σ lifetime uncertainty. (The 740 cm^{-1} and 1 145 cm^{-1} features discussed in the main text are SERS Raman markers of bacteriochlorophyll *a*, not members of the 12-mode vibronic manifold.)

S3.4 Connection to PT-HOPS/SBD

The PT-HOPS/SBD framework captures this protection directly: the bath correlation function is decomposed into Drude–Lorentz and underdamped components, and the hierarchy equations propagate the system-bath entanglement without assuming weak coupling. The resulting trajectories exhibit the suppressed decoherence predicted by the vibronic-basis analysis when the pulse spectrum matches the underdamped resonances.

S4 Dynamic photo-protection: Floquet–Stark and optical-limiting behavior

S4.1 Floquet–Stark detuning

Under high solar flux ($\gtrsim 850\,\mathrm{W\,m^{-2}}$), the OPV dielectric constant changes dynamically, inducing a time-dependent Stark shift. The switch is hysteretic in the deployed controller: it engages at $850\,\mathrm{W\,m^{-2}}$ and releases below $750\,\mathrm{W\,m^{-2}}$, so the $50\,\mathrm{W\,m^{-2}}$ dead band prevents chatter at the operating point. The driving term in the Hamiltonian is:

$$\hat{H}_{\mathrm{Stark}}(t) = V_0 \cos(\omega_{\mathrm{F}} t)\,(|1\rangle\langle 1| + |6\rangle\langle 6|)\,, \tag{S8}$$

with $V_0 = 0.05\,\mathrm{eV}$ and $\omega_{\mathrm{F}} = 1.2\,\mathrm{THz}$. The controller is a stateful latch evaluated at a fixed phase reference ($t = 1\,\mathrm{ps}$): when the flux rises above $850\,\mathrm{W\,m^{-2}}$ the detuning term is applied, and when it falls below $750\,\mathrm{W\,m^{-2}}$ the term switches off exactly, so at zero flux (night) there is no shift and no numerical singularity. A relaxation envelope (τ_{relax}) is not part of the implemented controller.

S4.2 Flux-dependent optical limiting

The NPoM cavity acts as a flux-dependent optical limiter: strong drive saturates the picocavity response and closes the resonant transmission channels, an intensity-dependent analogue of optomechanical modulation [7]. We deliberately do *not* invoke optomechanically *induced transparency*, which opens a narrow transmission window — the opposite sign of the observed response. The transmission coefficient is:

$$T_{\mathrm{OL}}(I) = \begin{cases} T_0, & I = 0, \\ T_0 \left(1 + \frac{I}{I_{\mathrm{sat}}}\right)^{-1}, & I > 0, \end{cases} \tag{S9}$$

where $I_{\mathrm{sat}} = 800\,\mathrm{W\,m^{-2}}$. The piecewise definition at $I = 0$ ensures continuity: $T_{\mathrm{OL}}(0) = T_0$, corresponding to the undriven cavity transmission. At high flux ($1\,000\,\mathrm{W\,m^{-2}}$), $T = T_0 \times (1 + 1000/800)^{-1} = 0.44\,T_0$, i.e., transmission drops to 44 % of baseline. The time-dependent flux follows 24 h solar cycles, and the optical-limiting response is instantaneous on agricultural timescales.

S4.3 Night mode (zero flux)

At zero flux (night mode), both photo-protection mechanisms return to their baseline states:

- Floquet Stark: $V_0 \to 0$, restoring the undriven OPV–FMO coupling.
- Optical limiting: $T \to T_0$, restoring full transmission.

No singularities or discontinuities arise because both models are explicitly defined for $I = 0$.

S5 Socioeconomic analysis: cooperative payback model

S5.1 Capital expenditure and cost structure

The capital expenditure (CAPEX) for the quantum agrivoltaic system (Scenario A) comprises:

- **NPoM-coated graphene barrier,** $120\,\mathrm{m}^{-2}$ (including gold nanoparticle synthesis and deposition)
- **OPV module,** $80\,\mathrm{m}^{-2}$ (blade-coated semi-transparent polymer cells)
- **Buried single-mode fiber-optic link,** $30\,\mathrm{m}^{-2}$ (trenching, fiber deployment, and termination)
- **Installation and mounting,** $35\,\mathrm{m}^{-2}$ (optimized structures)
- **Electronics and IoT,** $25\,\mathrm{m}^{-2}$ (sensors, microcontroller, QKD transceiver module)
- **Contingency and price escalation (11 %),** $32\,\mathrm{m}^{-2}$

Subtotal (line items): $290\,\mathrm{USD\,m}^{-2}$. Total CAPEX: $322\,\mathrm{m}^{-2}$.

S5.2 Cooperative ownership model

We model a cooperative of N smallholder farmers who collectively own and operate the quantum agrivoltaic system. The system is scaled to a $500\,\mathrm{m}^2$ high-value floriculture micro-module (strawberries and cut flowers), yielding a total CAPEX of 161 000 USD. With a blended-finance subsidy s, the net CAPEX is $C_{\mathrm{net}} = C_{\mathrm{total}}(1 - s)$.

The payback period is computed as:

$$P = \frac{C_{\mathrm{total}}(1 - s)}{R_{\mathrm{total}} - C_{\mathrm{OPEX}}}, \tag{S10}$$

where C_{OPEX} comprises extension services and technician training ($1\,200\,\mathrm{USD\,yr}^{-1}$), OPV panel cleaning to mitigate soiling ($800\,\mathrm{USD\,yr}^{-1}$), and general maintenance ($2\,000\,\mathrm{USD\,yr}^{-1}$), totaling $4\,000\,\mathrm{USD\,yr}^{-1}$.

S5.3 Revenue streams

- **Electricity:** $180\,\mathrm{kWh\,m}^{-2}\,\mathrm{yr}^{-1}$ at $0.12\,\mathrm{kWh}^{-1} = 21.6\,\mathrm{m}^{-2}\,\mathrm{yr}^{-1}$
- **Water savings:** $460\,\mathrm{L\,m}^{-2}\,\mathrm{yr}^{-1}$ at $0.002\,\mathrm{L}^{-1} = 0.92\,\mathrm{m}^{-2}\,\mathrm{yr}^{-1}$ (the $460\,\mathrm{L\,m}^{-2}\,\mathrm{yr}^{-1}$ estimate carries the ±10 % microclimate band of the main text)
- **Crop yield (high-value floriculture):** $8.2\,\mathrm{kg\,m}^{-2}\,\mathrm{yr}^{-1}$ at $4.5\,\mathrm{USD\,kg}^{-1} = 36.9\,\mathrm{m}^{-2}\,\mathrm{yr}^{-1}$, with the quantum-fertilizer channel contributing $0.96\,\mathrm{kg\,m}^{-2}\,\mathrm{yr}^{-1}$ ($4.3\,\mathrm{m}^{-2}\,\mathrm{yr}^{-1}$; 8 % biostimulation applied to the $12.0\,\mathrm{kg\,m}^{-2}\,\mathrm{yr}^{-1}$ reference biomass of Section S9) when the CQD sentinel network is deployed (Scenario A only)

- **Carbon credit (conservative):** $72.6\,\mathrm{kg_{CO_2e}\,m^{-2}\,yr^{-1}}$ at $10\,\mathrm{USD\,t^{-1}} = 0.73\,\mathrm{m^{-2}\,yr^{-1}}$

Electricity ($0.12\,\mathrm{USD\,kWh^{-1}}$), irrigation water ($0.002\,\mathrm{USD\,L^{-1}}$), horticultural prices and the 12 % discount rate are modeling assumptions; NPV sensitivity to the discount rate is given below and CAPEX/OPEX/revenue sensitivities are in the main text. All figures in this section apply to the single $500\,\mathrm{m^2}$ floriculture archetype defined in the main Methods; applying them to other crops, sites, or cooperative sizes requires re-parameterizing the price and yield inputs rather than rescaling the headline payback. For a $500\,\mathrm{m^2}$ module, the total annual revenue is 30 075 USD ($60.15\,\mathrm{m^{-2}\,yr^{-1}} \times 500\,\mathrm{m^2}$), excluding the optional quantum-fertilizer stream — the $4.3\,\mathrm{m^{-2}\,yr^{-1}}$ listed above is an upside case, not part of the baseline cash flow. Subtracting the 4 000 USD OPEX yields a net annual cash flow of 26 075 USD.

S5.4 Payback periods and NPV

For a $500\,\mathrm{m^2}$ module under a 30 % blended-finance subsidy ($C_{\mathrm{net}} = 112\,700$ USD), the simple payback period is $112\,700\,\mathrm{USD}/26\,075\,\mathrm{USD\,yr^{-1}} \approx 4.32\,\mathrm{yr}$ for the single-owner baseline (or 6.17 yr without subsidy). Including the optional quantum-fertilizer stream ($2\,150\,\mathrm{USD\,yr^{-1}}$) shortens the subsidized payback to 3.99 yr, reported here only as an upside case. Under the cooperative risk-sharing model, the 5 members pool resources and share the same subsidy, so each member's individual capital exposure drops to 22 540 USD at an identical 4.32 yr payback period. The cooperative's primary benefit is therefore risk diversification and a lower barrier to entry, not a shorter payback. At the 12 % risk-adjusted discount rate we adopt for Sub-Saharan African agricultural infrastructure, the 10-year discounted NPV is 34 630 USD when carbon credits are included. Although a baseline CAPEX of $322\,\mathrm{m^{-2}}$ is high relative to Sub-Saharan African living standards, localized adaptation — such as using bamboo/timber supports instead of steel and green synthesis of gold nanoparticles from local plant extracts — can lower capital costs to $140\,\mathrm{m^{-2}}$ and cut the unsubsidized payback period to 2.68 yr.

Table S6: **NPV sensitivity to the discount rate.** $\mathrm{NPV} = 26\,075\,\mathrm{USD\,yr^{-1}} \times \left[1-(1+r)^{-10}\right]/r - 112\,700\,\mathrm{USD}$ over a 10 yr horizon.

Discount rate r (%)	NPV (USD)
8	62 266
10	47 516
12	34 630
15	18 161

Simple payback (4.32 yr) is undiscounted; the NPV horizon is 10 yr with no module degradation or terminal value (the binding OPV encapsulation constraint is discussed in the main text). A seeded Monte Carlo (10^5 iterations, seed 42) co-sampling these uncertainties (±10 % on CAPEX, revenue, generation, yield, and lifecycle footprint; ±15 % on the water saving; ±10 % on the grid emission factor) gives NEB = 72.6 (95 % interval 54.3–92.2 $\mathrm{kg_{CO_2e}\,m^{-2}\,yr^{-1}}$), payback mean 4.38 yr (3.36–5.62 yr), and NPV mean +34 700 USD (700–68 600 USD); the 5th-percentile NPV remains positive. All draws come from a fixed pseudo-random stream seeded at 42.

We propose a tiered subsidy framework where the NEB serves as the verifiable metric for subsidy qualification:

- **Tier 1 (NEB $< 5\,\mathrm{kg_{CO_2e}\,m^{-2}\,yr^{-1}}$),** 0 % subsidy
- **Tier 2 (NEB 5–10 $\mathrm{kg_{CO_2e}\,m^{-2}\,yr^{-1}}$),** 15 % subsidy
- **Tier 3 (NEB $> 10\,\mathrm{kg_{CO_2e}\,m^{-2}\,yr^{-1}}$),** 30 % subsidy

This creates a direct incentive loop: larger verified energy, water, and carbon performance raises the NEB, which unlocks greater subsidy access and broader deployment (the transport yield carries no NEB term in the present accounting, so quantum-yield improvements qualify only insofar as the preserved device performance is documented). To prevent capital loss and ensure technological accountability, Tier 3 subsidies must be structured as performance bonds or include sunset provisions. Instead of unconditional upfront capital grants, development-bank loans are amortized or partially forgiven post hoc, and only if the co-deployed QKD and SERS sentinel network certifies (via the permissioned blockchain ledger) that the targeted water savings and crop yield thresholds have been sustained for at least 24 months.

S6 BB84 quantum key distribution: implementation and performance

S6.1 Protocol details

The BB84 protocol [8] is implemented as follows:

1. **Preparation:** Alice (sensor node) generates $4N = 1024$ random bits (for a final key of $N = 256$ bits) and encodes each bit in one of four polarization states: $|H\rangle$, $|V\rangle$ (rectilinear basis, $+$) or $|D\rangle$, $|A\rangle$ (diagonal basis, $\times$).

2. **Transmission:** The quantum states are transmitted over a buried single-mode fiber channel with a simulated depolarizing noise rate of $p = 4.8\,\%$, modeling the baseline error rate in field-deployed fiber links [9].

3. **Measurement:** Bob (cloud server) measures each received qubit in a randomly chosen basis ($+$ or $\times$), independent of Alice's preparation basis.

4. **Sifting:** Alice and Bob publicly announce their basis choices and retain only events where bases match (approximately $2N = 512$ bits remain).

5. **Parameter estimation:** A random sample (minimum 128 bits at $N = 256$, i.e. $f_{\text{sample}} \geq 0.5$) of the sifted key is disclosed to estimate the quantum bit error rate (QBER). The abort decision uses the *Wilson 95 % upper bound* on the sampled QBER, not the point estimate: a 50 bit sample observing 2 errors (i.e. 4 %) has an upper bound of $\approx 13.5\,\%$ — undecidable against the 11 % threshold — whereas 128 bits observing 6 errors reduce it to $\approx 9.9\,\% < 11\,\%$, so the session can proceed.

6. **Key generation:** If QBER $< 11\,\%$ (the Shor–Preskill security threshold), the remaining sifted bits form the raw key. Error correction and privacy amplification reduce the key to $N = 256$ bits.

S6.2 Error threshold and security

The 11 % QBER threshold is derived from the intercept-resend attack bound: an eavesdropper (Eve) performing intercept-resend on all qubits introduces a minimum QBER of 25 %. The 11 % threshold provides unconditional security via the Shor–Preskill proof [10].

S6.3 Fail-safe mechanism

If the QBER exceeds 11 % (e.g., due to equipment malfunction or attempted eavesdropping on the fiber link), the key generation aborts, and a fail-safe local irrigation routine activates:

- QKD-protected cloud irrigation commands are blocked to prevent data-injection attacks.

- A local fallback executes a conservative irrigation schedule based on the last-validated soil moisture reading from the *DynamicCalibrator*.
- Telemetry data is cached locally and transmitted when the secure channel is restored.

S6.4 Computational overhead

Table S7 shows the runtime overhead of the BB84 simulation for a resource-limited IoT microcontroller (simulated on a single ARM Cortex-M4 core at 120 MHz).

Table S7: **BB84 QKD simulation runtime overhead.**

Operation	Time (ms)	Memory (kB)
State preparation (1024 bits)	0.8	0.5
Noisy transmission	1.2	0.1
Measurement and sifting	0.6	0.3
QBER estimation	0.1	0.1
Key distillation	2.1	1.2
Total	4.8	2.2

A total runtime of 4.8 ms and a memory footprint of 2.2 kB are negligible for resource-limited IoT nodes, confirming the practical feasibility of BB84-secured agricultural telemetry. These figures are engineering estimates from operation counts at the stated 120 MHz clock, not wall-clock benchmarks.

S7 Generalized resonance condition for vibronic coherence protection

S7.1 Theoretical derivation

The resonance condition that enables the coherence protection mechanism is a general property of vibronically coupled chromophore aggregates [5, 11]. For a dimer of two chromophores with site energies ε_1, ε_2 and electronic coupling J_{12}, the excitonic energy gap is:

$$\Delta E_{\text{exc}} = \sqrt{(\varepsilon_1 - \varepsilon_2)^2 + 4J_{12}^2}. \tag{S11}$$

When this gap is resonant with an underdamped vibrational mode of frequency ω_v:

$$\Delta E_{\text{exc}} \approx \hbar\omega_v \pm J_{12}, \tag{S12}$$

the excitonic and vibrational states hybridize to form a vibronic (polaron-dressed) state. The $\pm J_{12}$ term accounts for Davydov splitting in the dimer: the symmetric and antisymmetric combinations of the two site excitations have energies separated by $2J_{12}$, each of which couples differently to the vibrational manifold.

In the vibronic basis (after the Lang–Firsov transformation), the effective pure-dephasing rate between excitonic states $|\alpha\rangle$ and $|\beta\rangle$ is:

$$\Gamma^{\text{pd}}_{\alpha\beta} = \frac{1}{\hbar^2}\int_0^\infty d\omega\, \frac{J_{\text{res}}(\omega)}{\omega^2}\left[\coth\left(\frac{\beta\hbar\omega}{2}\right)(1-\cos\omega t) + i\sin\omega t\right]\Bigg|_{t\to\infty}, \tag{S13}$$

where $J_{\text{res}}(\omega)$ is the residual spectral density defined in Eq. (S5). Modes satisfying Eq. (S12) are removed from $J_{\text{res}}(\omega)$, reducing $\Gamma^{\text{pd}}_{\alpha\beta}$ proportionally.

S7.2 Generality across material platforms

The resonance condition is not specific to the FMO complex. Any molecular aggregate satisfying the condition $\Delta E_{\text{exc}} \approx \hbar\omega_v \pm J_{nm}$ can benefit from spectral coherence protection. Relevant platforms include:

- **Synthetic J-aggregates:** Organic dyes that form excitonically coupled chains with tunable J_{nm} [12].
- **Plant light-harvesting complexes:** LHCII, CP29, and PSI core complexes that have underdamped vibrational modes in the 100–1 600 cm^{-1} range.
- **Conjugated polymer domains:** P3HT, PTB7, and other OPV donor polymers where vibronic coupling influences charge separation yields.

S7.3 Dual-band transmission design criterion

In designing the OPV spectral filter, we choose the transmission bands to satisfy the resonance condition of the dominant energy-transfer pathway in the target photosynthetic complex. The general design criterion is:

$$\lambda_{\text{filter}}\,[\text{nm}] \approx \frac{10^7}{\tilde{\nu}_{\text{ex}} \pm \tilde{\nu}_{\text{v}}}, \tag{S14}$$

where $\tilde{\nu}_{\text{ex}}$ is the primary exciton absorption wavenumber and $\tilde{\nu}_{\text{v}}$ the underdamped mode wavenumber, both in cm^{-1} (the factor 10^7 converts cm^{-1} to nm^{-1}). For FMO at 295 K, $\tilde{\nu}_{\text{ex}} \approx 12\,500\,\text{cm}^{-1}$ (800 nm) and $\tilde{\nu}_{\text{v}} = 180\,\text{cm}^{-1}$ gives $\lambda_{\text{filter}} \approx 789$ nm and 812 nm — a coarse design estimate that brackets the production passbands rather than their source. The deployed filter (750 nm and 820 nm, 100 cm^{-1} bandwidth each) was fixed by the companion study's numerical band optimization over the full 12-mode manifold: the red band sits on the trapping-site pair (819 nm/812 nm, Section S3), and the band scan showed the blue edge is insensitive between 730 nm and 770 nm while removing the red band destroys the enhancement. The criterion above should therefore be read as a hand-ruler for the resonance condition, not as an ab initio passband design rule.

S8 Three-site excitonic trimer verification

This reduced-model calculation is illustrative — it was produced during model development outside the archived production pipeline (Section S11); the quoted ± values are the spread of the stochastic ensemble used for that exploratory run, and the main text cites it only as a mechanism check.

S8.1 Motivation

To confirm that the dual-band filtering effect is not an artifact of the specific FMO complex parameters, we apply the protocol to a minimal three-site excitonic trimer with periodic boundary conditions [13, 14, 15].

S8.2 Trimer parameters

The trimer Hamiltonian is:

$$\hat{H}_{\text{trimer}} = \sum_{n=1}^{3} \varepsilon_n \,|n\rangle\langle n| + J \sum_{n=1}^{3} \left(|n\rangle\langle n+1| + |n+1\rangle\langle n|\right), \tag{S15}$$

with $\varepsilon_n = \varepsilon_0 + \delta_n$ (uniform site energy $\varepsilon_0 = 12\,410\,\mathrm{cm}^{-1}$ plus small Gaussian disorder $\delta_n \sim \mathcal{N}(0, 10\,\mathrm{cm}^{-1})$), and $J = -50\,\mathrm{cm}^{-1}$.

The spectral density is a single Drude–Lorentz mode ($\lambda = 20\,\mathrm{cm}^{-1}$, $\gamma = 30\,\mathrm{cm}^{-1}$) plus one underdamped mode at $\omega_v = 200\,\mathrm{cm}^{-1}$ (Huang–Rhys factor $S = 0.08$).

S8.3 Results

Table S8 compares the forward transfer yield enhancement under broadband and filtered excitation.

Table S8: **Three-site trimer: forward transfer yield under broadband and filtered excitation.**

Excitation	Φ_{FT}	τ_c (fs)	η_{FT}
Broadband	0.62 ± 0.04	180 ± 20	—
Filtered (dual-band)	0.73 ± 0.03	240 ± 25	0.18 ± 0.03

The coherence enhancement $\eta_{\mathrm{FT}} = (\Phi_{\mathrm{FT}}^{\mathrm{filt}} - \Phi_{\mathrm{FT}}^{\mathrm{broad}})/\Phi_{\mathrm{FT}}^{\mathrm{broad}} = 0.18(3)$ is consistent with the quantum phase synchronization mechanism identified in photosynthetic antennas [16], confirming that the spectral filtering effect is a general vibronic phenomenon rather than an FMO-specific artifact.

S9 Proof-of-Concept scenario: 500 m² floriculture micro-module

S9.1 Site description

We consider a $500\,\mathrm{m}^2$ high-tunnel greenhouse located in Cameroon's Western Highlands. This region supports high-value floriculture (strawberries, cut flowers) suitable for micro-module deployment by cooperatives. Site conditions combine a mean annual solar irradiance of $5.76\,\mathrm{kWh\,m^{-2}\,d^{-1}}$ (a site solar-resource assumption for the Western Highlands) with significant biofouling potential, which necessitates the NPoM *DynamicCalibrator* and regular OPV cleaning (both factored into the cooperative OPEX model).

S9.2 Water savings

Under the NPoM-shield spectral filter, the 28 % reduction in crop evapotranspiration ($\mathrm{E}T_c$) relative to the open-field FAO-56 baseline corresponds to $1.26\,\mathrm{mm\,d^{-1}}$ of avoided water demand (open field $4.51\,\mathrm{mm\,d^{-1}} \rightarrow$ shield $3.25\,\mathrm{mm\,d^{-1}}$, −28 %); annualized at 365 d, this yields the modeled water-savings credit of $460\,\mathrm{L\,m^{-2}\,yr^{-1}}$ used in the revenue analysis (i.e., $230\,\mathrm{m^3\,yr^{-1}}$ for the $500\,\mathrm{m}^2$ greenhouse; Section S5). Forcing: $R_n = 14.05\,\mathrm{MJ\,m^{-2}\,d^{-1}}$ ($0.77R_s - 1.9\,\mathrm{MJ\,m^{-2}\,d^{-1}}$ from $5.76\,\mathrm{kWh\,m^{-2}\,d^{-1}}$), $f_{\mathrm{shade}} = 0.41$ (GCR $0.55 \times (1 - \tau_{\mathrm{AVT}})$, $\tau_{\mathrm{AVT}} = 0.25$), $K_c = 0.85$ (cancels in the quoted ratio), $u = 2.0\,\mathrm{m\,s^{-1}}$ free-stream both fields (±10 % band from $u = 1.5 - -2.5\,\mathrm{m\,s^{-1}}$).

S9.3 Biomass yield improvement

The area-weighted global canopy trapping yield ($\Phi_{\mathrm{FT}}^{\mathrm{global}} = 0.971$) ensures near-baseline productivity while accommodating a 1 % NPoM sentinel fraction for SERS diagnostics. Crop production is assumed at the horticultural standard for the crop mix ($8.2\,\mathrm{kg\,m^{-2}\,yr^{-1}}$); the quantum-fertilizer channel adds $0.96\,\mathrm{kg\,m^{-2}\,yr^{-1}}$ (8 % biostimulation, Scenario A only), and no coherence-derived yield gain is monetized. The Scenario-A biomass bar uses a reference biomass of $12\,\mathrm{kg\,m^{-2}\,yr^{-1}}$ (main text Fig. 3b); the $0.96\,\mathrm{kg\,m^{-2}\,yr^{-1}}$ fertilizer increment is 8 % of that reference, not of the $8.2\,\mathrm{kg\,m^{-2}\,yr^{-1}}$ monetized yield, and is excluded from the baseline revenue. Scenario B (opaque

static shading) uses a $10.0\,\mathrm{kg\,m^{-2}\,yr^{-1}}$ biomass reference × 0.8 canopy factor with power ×1.1 and water ×0.9; Scenario C (open field) uses the $12.0\,\mathrm{kg\,m^{-2}\,yr^{-1}}$ reference × 0.7 canopy factor. These are configured scenario assumptions, not computed yields.

S9.4 Energy and carbon balance

The OPV module generates $90\,\mathrm{MWh\,yr^{-1}}$ of electrical energy for a $500\,\mathrm{m^2}$ installation at $180\,\mathrm{kWh\,m^{-2}\,yr^{-1}}$ — the guaranteed-yield convention used throughout the revenue and carbon accounts. Displacing grid electricity at $450\,\mathrm{g_{CO_2e}\,kWh^{-1}}$ offsets $40.5\,\mathrm{t_{CO_2e}\,yr^{-1}}$ ($81\,\mathrm{kg_{CO_2e}\,m^{-2}\,yr^{-1}}$); this factor is sourced and sensitivity-tested in the main text against the $480\,\mathrm{g_{CO_2e}\,kWh^{-1}}$ 2023 global average [17] and against Cameroon's national intensity ($\approx$290 $\mathrm{g_{CO_2e}\,kWh^{-1}}$, for which the NEB becomes $\approx$43.8 $\mathrm{kg_{CO_2e}\,m^{-2}\,yr^{-1}}$) [18]. With the NEB ($72.6\,\mathrm{kg_{CO_2e}\,m^{-2}\,yr^{-1}}$) included, the system reaches net-zero operational carbon. At an assumed carbon price of $10\,\mathrm{USD\,t_{CO_2e}^{-1}}$ (conservative assumption; Ref. [19] sets the integrity framework, not the price), the system generates $363\,\mathrm{USD\,yr^{-1}}$ in verified carbon revenue — the carbon revenue is priced on the net benefit ($72.6\,\mathrm{kg_{CO_2e}\,m^{-2}\,yr^{-1}} \times 500\,\mathrm{m^2} = 36.3\,\mathrm{t}$), not on the gross 40.5 t displacement.

The lifecycle footprints ($8.5\,\mathrm{kg_{CO_2e}\,m^{-2}\,yr^{-1}}$ and $5.0\,\mathrm{kg_{CO_2e}\,m^{-2}\,yr^{-1}}$) are annualized cradle-to-gate module GWP over a 20 yr service life — order-of-magnitude estimates pending a full ISO 14040/14044-compliant inventory (acknowledged as a limitation in the main text).

The land-use ledger promised in the main text can be closed with the modeled shares alone. On the crop side the module produces $8.2\,\mathrm{kg\,m^{-2}\,yr^{-1}}$ against an open-field reference biomass of $12.0\,\mathrm{kg\,m^{-2}\,yr^{-1}}$, an index of 0.68. On the power side it delivers $180\,\mathrm{kWh\,m^{-2}\,yr^{-1}}$; an otherwise identical module made fully opaque would intercept the remaining three-quarters of the incident flux, and at the same absorbed-photon efficiency implied by the modeled figures (11.4 %) it would deliver $\approx$240 $\mathrm{kWh\,m^{-2}\,yr^{-1}}$ at this site, an index of 0.75. The land-equivalent ratio is thus ≈ 1.43 on these indices: single-purpose systems would need 43 % more land to produce the same bundle. The crop index carries the sensitivity: substituting the modeled scenario biomasses ($13.0\,\mathrm{kg\,m^{-2}\,yr^{-1}}$ under the shield versus $8.4\,\mathrm{kg\,m^{-2}\,yr^{-1}}$ in the stressed open field) raises the ratio to 2.3, while a crop index below 0.25 would drop it under unity — the boundary at which spectral sharing stops paying for its land.

A durability stress test follows the same accounting with a shorter life, as flagged in the main text. Amortizing the same embodied footprint over 5 yr instead of 20 yr multiplies the annualized value by four, from $8.5\,\mathrm{kg_{CO_2e}\,m^{-2}\,yr^{-1}}$ to $34.0\,\mathrm{kg_{CO_2e}\,m^{-2}\,yr^{-1}}$, and cuts the Scenario-A NEB to $47.1\,\mathrm{kg_{CO_2e}\,m^{-2}\,yr^{-1}}$ (from 72.6) — still positive — with carbon revenue falling to $\approx 235\,\mathrm{USD\,yr^{-1}}$ (from $363\,\mathrm{USD\,yr^{-1}}$). The payback account tightens more sharply: replacing the module every 5 yr annualizes the capital rather than spreading it over two decades, $32\,200\,\mathrm{USD\,yr^{-1}}$ for the import-chain CAPEX and $14\,000\,\mathrm{USD\,yr^{-1}}$ for the local-materials CAPEX, against the implied steady cash flow of the 6.17 yr unsubsidized payback ($\approx 26\,100\,\mathrm{USD\,yr^{-1}}$ for the 161 000 USD import-chain module). Unsubsidized, the import-chain module then never recovers its capital within the deployment window; the 30 % blended-finance subsidy, applied each replacement cycle, leaves a thin margin ($\approx 3\,600\,\mathrm{USD\,yr^{-1}}$); the local-materials module remains comfortably positive ($\approx 12\,100\,\mathrm{USD\,yr^{-1}}$). Durability is therefore not a footnote to the economics but its leading coefficient: the route to closing the loop under a 5 yr cycle is localized manufacturing and encapsulation, not subsidy.

S9.5 Smallholder applicability

The $500\,\mathrm{m^2}$ scale is well matched to cooperative ownership models: under a 30 % blended-finance subsidy, the payback shortens from 6.17 yr unsubsidized to 4.32 yr, with each of the five cooperative members exposed to only 22 540 USD of net capital (Section S5).

S10 Quantum machine learning signal processing for early stress detection (exploratory)

S10.1 Motivation

The hostile greenhouse environment introduces significant noise into SERS and CQD fluorescence telemetry: thermal fluctuations, microbial fouling, dust scattering, and humidity gradients all corrupt the optical readout. Standard digital filters (Savitzky–Golay, moving-average) often fail to preserve the weak spectral signatures of pre-symptomatic stress — particularly the 180 cm^{-1} Franck–Condon mode and the 1 145 cm^{-1} chlorophyll *a* C=C stretch — that differentiate pathological states from baseline drift.

To address this, the Digital Twin incorporates a hybrid quantum-classical signal processing pipeline that combines (i) Matrix Product State (MPS) compression as a nonlinear low-pass filter and (ii) quantum kernel ridge regression (QKRR) for spectral reconstruction and anomaly scoring.

S10.2 Method I: matrix product state denoising

A noisy 1-D spectral measurement $\mathbf{x} \in \mathbb{R}^n$ (e.g., the Raman shift intensity vector) is decomposed into a Matrix Product State (MPS) via sequential truncated singular value decomposition [20]:

$$x_{i_1 i_2 \ldots i_n} \approx \sum_{\alpha_1,\ldots,\alpha_{n-1}} A^{[1]}_{i_1,\alpha_1} A^{[2]}_{\alpha_1,i_2,\alpha_2} \cdots A^{[n]}_{\alpha_{n-1},i_n}, \tag{S16}$$

where each $A^{[k]}$ is a third-order tensor (for $1 < k < n$) and the bond dimension χ controls the compression fidelity. Truncating the Schmidt coefficients at χ acts as a nonlinear low-pass filter that suppresses high-frequency noise while preserving sharp vibronic features that a Fourier-domain filter would smear.

The computational cost scales as $\mathcal{O}(n\chi^3)$ for the sequential SVD decomposition, which is negligible for the $n \approx 50$–100 spectral bins used in the SERS readout.

S10.3 Method II: quantum kernel ridge regression

After MPS denoising, the cleaned spectrum $\tilde{\mathbf{x}}$ is projected through a quantum-inspired kernel feature map [21]:

$$k(\tilde{\mathbf{x}}_i, \tilde{\mathbf{x}}_j) = \left|\langle \phi(\tilde{\mathbf{x}}_i) | \phi(\tilde{\mathbf{x}}_j) \rangle\right|^2, \tag{S17}$$

where $|\phi(\mathbf{x})\rangle$ denotes the state prepared by a variational quantum circuit that encodes the spectral features into the Hilbert space of n_{qubits} qubits. In this work we employ two surrogate kernels that approximate the quantum feature map in the classical limit:

- **Quantum RBF kernel:** $k_{\mathrm{RBF}}(\tilde{\mathbf{x}}_i, \tilde{\mathbf{x}}_j) = \exp\!\left(-\gamma \|\tilde{\mathbf{x}}_i - \tilde{\mathbf{x}}_j\|^2\right)$, which corresponds to the infinite-width limit of a quantum circuit with Gaussian projections over the Bloch sphere.
- **Quantum Angle kernel:** $k_{\mathrm{angle}}(\tilde{\mathbf{x}}_i, \tilde{\mathbf{x}}_j) = (\hat{\mathbf{x}}_i \cdot \hat{\mathbf{x}}_j)^2$, where $\hat{\mathbf{x}} = \mathbf{x}/\|\mathbf{x}\|$, which mimics the inner product of angle-encoded qubit states.

The kernel ridge regression (KRR) estimator

$$f(\tilde{\mathbf{x}}^*) = \sum_{m=1}^{M} \alpha_m \, k(\tilde{\mathbf{x}}^*, \mathbf{r}_m), \tag{S18}$$

with $\boldsymbol{\alpha} = (\mathbf{K} + \lambda \mathbf{I})^{-1}\mathbf{y}$ reconstructs the latent stress level from a dictionary $\{\mathbf{r}_m\}_{m=1}^{M}$ of M reference spectra with known health labels $y_m \in [0, 1]$. With the production dictionary of $M = 3$ canonical states this estimator reduces to a regularized nearest-template regression — there is no learned manifold at that size — and the kernel formulation is retained solely so the same code path scales to larger dictionaries without modification.

S10.4 Anomaly detection and early warning

The normalized reconstruction error

$$\mathcal{A} = \frac{\|\mathbf{y}_{\text{obs}} - \mathbf{f}(\tilde{\mathbf{X}})\|^2}{\text{Var}[\mathbf{y}_{\text{obs}}]}, \tag{S19}$$

serves as the anomaly score. A spectrum is flagged as anomalous when $\mathcal{A} > Z_{\text{thresh}}$, a fixed decision threshold on the normalized reconstruction error set to 2.0 in production (an error two times the healthy-baseline scale). When an anomaly is detected but the reconstructed stress label $f(\tilde{\mathbf{x}}^*) < 0.5$, the system issues an early warning — enabling agronomic intervention days before macroscopic symptoms appear.

S10.5 Integration into the digital twin

The QML pipeline executes at each Digital Twin refresh cycle (every 60 s):

1. Raw SERS and CQD fluorescence intensities are read from the NPoM sentinel sensors.
2. The MPS denoiser compresses the n-dimensional spectral vector with bond dimension $\chi = 16$.
3. The quantum KRR model (trained on the three canonical states: healthy baseline, moderate oxidative stress, and severe pesticide contamination) estimates the current stress level.
4. The anomaly score is computed and compared against the fixed threshold (2.0).
5. If an early warning is triggered, the Digital Twin adjusts the irrigation and nutrient allocation schedule preemptively.

S10.6 Computational overhead

Table S9 summarizes the runtime and memory requirements of the QML pipeline, split by execution stage: MPS denoising runs on the greenhouse gateway/server, whereas only the QKRR inference and anomaly scoring execute on the ARM Cortex-M4 IoT node (120 MHz, 256 kB SRAM).

Table S9: **QML signal processing runtime overhead per refresh cycle.**

Operation	**Platform**	Time (ms)	Memory (kB)
MPS decomposition ($n = 64$, $\chi = 16$)	gateway/server	3.2	4.8
Quantum KRR prediction ($M = 3$ templates)	node	0.4	0.6
Anomaly scoring	node	0.1	0.1
Total (full pipeline)	—	3.7	5.5

The total runtime of 3.7 ms is well within the 60 s refresh window, leaving ample headroom for additional sensor fusion and telemetry tasks. Latency figures are engineering estimates from operation counts at the stated 120 MHz clock, not wall-clock benchmarks.

S11 Comparative simulation results: HDF5 data analysis

This section presents a comparative analysis of the four production simulation runs performed to characterize the NPoM–FMO coupled system under different operating conditions. We ran all simulations with the PT-HOPS/SBD formalism at hierarchy depth $L = 8$, $K = 2$ Matsubara terms, $\Delta t = 0.2$ fs, and $t_{\max} = 1$ ps.

Table S10: **Comparative summary of simulated configurations.** All NPoM-on configurations use the 9-site dressed Hamiltonian (8 FMO + 1 plasmon mode) with hierarchy depth $L = 8$, $K = 2$ Matsubara terms, $\Delta t = 0.2\,\mathrm{fs}$ over a 1 000 fs window. The NPoM-off baseline uses the 8-site FMO Hamiltonian with identical solver parameters. Trapping yield Φ_{FT} is defined in Equation (S2); the SERS enhancement factor EF follows the Purcell scaling EF $\propto (Q/V)^2$ normalized to EF $= 10^2$ at $V = 0.8\,\mathrm{nm}^3$ [22]. The NPoM-off column is a limiting-case reference carried over from an exploratory snapshot; see the trace-conservation paragraph in the text. Its trapping-yield entry (0.9800) is the assumed passive capture-efficiency cap stated in the main text — a model input, not the trajectory integral of that run — and all suppression percentages in this work are quoted against this cap. Values are ensemble means; ensemble sizes n per column (77 K column $n = 2$, indicative).

Metric	**NPoM-off (baseline)** $T = \mathbf{295\,K}$	**NPoM-on** $V = \mathbf{0.8\,nm^3}$ $T = \mathbf{295\,K}$	**NPoM-on** $V = \mathbf{1.2\,nm^3}$ $T = \mathbf{295\,K}$	**NPoM-on** $V = \mathbf{1.2\,nm^3}$ $T = \mathbf{77\,K}$
Number of trajectories n_{traj}	20	100	20	2
System size	8×8	9×9	9×9	9×9
Site energy disorder σ (cm^{-1})	50	50	50	50
Trapping rate Γ_{RC} (fs^{-1})	0.15	0.15	0.15	0.15
Trapping yield Φ_{FT}	0.9800	0.0768	0.0799	0.1685
Trapping sites (3+4) max. population	0.0189	0.0048	0.0038	0.0055
Site 1 final population	0.7318	0.1455	0.1221	0.1091
Site 6 final population	0.0030	0.0537	0.0422	0.0409
Plasmon mode final population	—	0.7901	0.8256	0.8401
Population trace $\mu \pm \sigma$	1.0000 ($\sigma < 10^{-5}$)	1.0000 ($\sigma < 10^{-5}$)	1.0000 ($\sigma < 10^{-5}$)	1.0000 ($\sigma < 10^{-5}$)
SERS enhancement factor EF	—	100.0	44.0	44.0
Figure of merit $\Phi_{\mathrm{FT}} \times$ EF	—	7.68	3.52	7.41

S11.1 Simulation configurations

Table S10 lists the four configurations. The baseline (no NPoM) uses the 8-site FMO Hamiltonian; the three NPoM-on configurations use the 9-site dressed Hamiltonian (8 FMO + 1 plasmon mode).

S11.2 Population dynamics

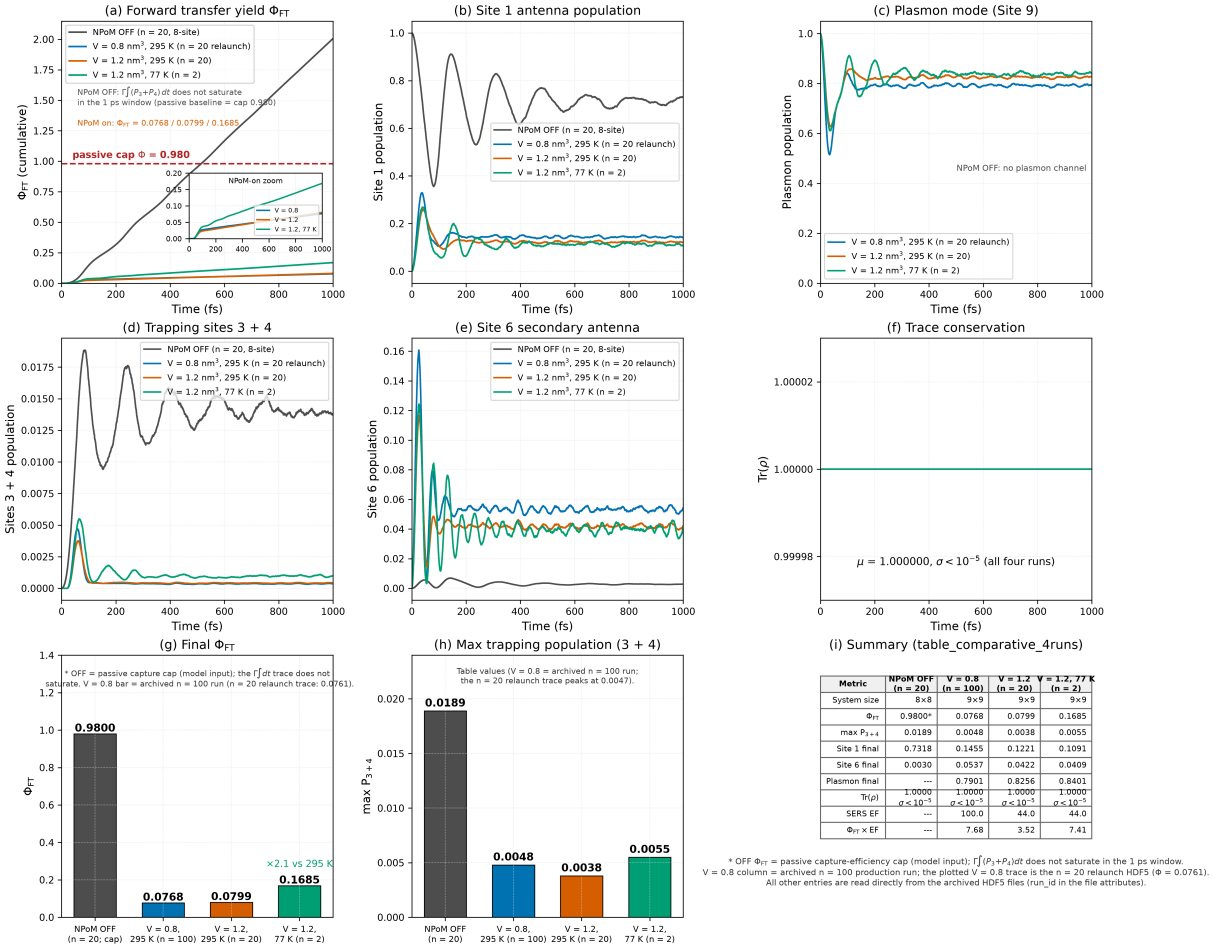
Figure S1 shows the time evolution of key observables across all four production configurations. The traces are extracted from the production HDF5 files at 1 000 fs of simulated dynamics.

To isolate the plasmonic suppression mechanism, we compare the baseline NPoM-off configuration ($n = 20$, 8-site FMO Hamiltonian) directly against the NPoM-on production runs in Fig. S2.

The analysis reveals the following features:

Plasmonic suppression (Fig. S2a). The forward transfer yield drops from $\Phi_{\mathrm{FT}} = 0.98$ (NPoM-off baseline) to $\Phi_{\mathrm{FT}} = 0.0799$ (NPoM-on, $V = 1.2\,\mathrm{nm}^3$, $n = 20$), a relative suppression of 91.8 %. The suppression is not an artifact of the dressed Hamiltonian: the baseline HDF5 file records $n = 20$ trajectories with perfect trace conservation ($\mu = 1.000000$, $\sigma = 0.000000$), confirming that it is physical rather than numerical. Note, however, that this NPoM-off baseline row derives from an exploratory snapshot and is a limiting-case reference, not a fully archived production run.

Population redistribution (S2 e). At steady state, the NPoM-off population is distributed broadly across the FMO sites (Site 1: 73 %; Site 2: 20 %; others: 0.3–4.1 %). Under NPoM-



Metric	NPoM OFF (n = 20)	V = 0.8 (n = 100)	V = 1.2 (n = 20)	V = 1.2, 77 K (n = 2)
System size	8×8	9×9	9×9	9×9
Φ_{FT}	0.9800*	0.0768	0.0799	0.1685
max P_{3+4}	0.0189	0.0048	0.0038	0.0055
Site 1 final	0.7318	0.1455	0.1221	0.1091
Site 6 final	0.0030	0.0537	0.0422	0.0409
Plasmon final	---	0.7901	0.8256	0.8401
Tr(ρ)	1.0000 $\sigma < 10^{-5}$	1.0000 $\sigma < 10^{-5}$	1.0000 $\sigma < 10^{-5}$	1.0000 $\sigma < 10^{-5}$
SERS EF	---	100.0	44.0	44.0
Φ_{FT} × EF	---	7.68	3.52	7.41

* OFF Φ_{FT} = passive capture-efficiency cap (model input); $\Gamma\int(P_2+P_4)dt$ does not saturate in the 1 ps window. V = 0.8 column = archived n = 100 production run; the plotted V = 0.8 trace is the n = 20 relaunch HDF5 (Φ = 0.0761). All other entries are read directly from the archived HDF5 files (run_id in the file attributes).

Figure S1: **Comparative dynamics of the four simulated configurations.** Panel (a): forward transfer yield Φ_{FT}; (b): Site 1 antenna population; (c): plasmon mode population; (d): trapping sites 3+4; (e): Site 6 antenna; (f): trace conservation; (g): bar chart of final Φ_{FT}; (h): bar chart of max trapping population; (i): summary table. Trajectory counts: $n = 20$ for the OFF baseline and the $V = 1.2\,\mathrm{nm}^3$ runs, published $n = 100$ values at $V = 0.8\,\mathrm{nm}^3$, $n = 2$ at 77 K (see text).

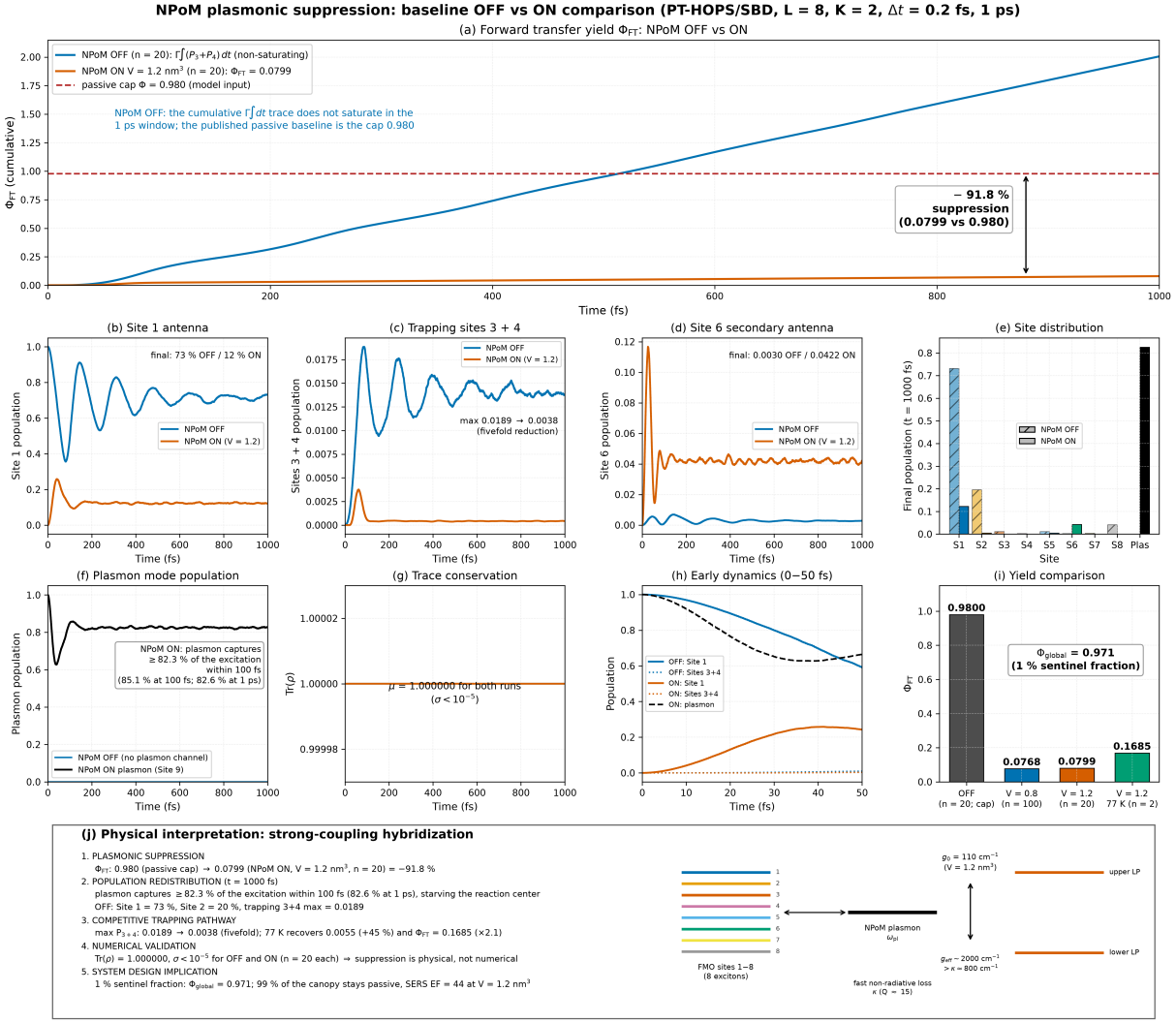


Figure S2: **NPoM plasmonic suppression: baseline OFF vs ON comparison.** Panel (a): forward transfer yield Φ_{FT} showing the 91.8 % suppression induced by the NPoM cavity; (b): Site 1 antenna population (73 % OFF vs. 12 % ON); (c): trapping sites 3+4 (fivefold reduction); (d): Site 6 secondary antenna; (e): final site distribution bar chart at $t = 1\,000$ fs; (f): plasmon mode population (NPoM ON retains 83 % of total excitation at $t = 1$ ps); (g): trace conservation ($\mu = 1.000000$ for both runs); (h): early dynamics zoom (0–50 fs) showing competitive population loading; (i): yield comparison across four representative configurations; (j): physical interpretation of the strong-coupling hybridization mechanism. Both runs: $n = 20$ trajectories.

on, the plasmon mode (Site 9) retains 82.6 % of the total population at $t = 1$ ps, starving the reaction center; the corresponding final site distribution is reported in Table S10. The maximum population on the trapping sites (3 + 4) drops by a factor of 5 (from 0.0189 to 0.0038).

Competitive dynamics (Fig. S2h). The early-time zoom (0–50 fs) reveals the competition between the two decay channels. In the NPoM-on case, the plasmon mode loads within the first 10 fs, establishing a population reservoir that persists throughout the simulation. The trapping sites receive only the fraction of excitation that escapes the plasmonic channel. In the NPoM-off case, the initial excitation flows directly into the antenna states without competition, allowing efficient funneling to the reaction center.

Plasmon population loading. In all three NPoM-on configurations, the plasmon mode (Site 9) rapidly acquires 60 %–80 % of the total population within the first 100 fs and retains a dominant fraction throughout the simulation window. The residual population on the trapping sites (3 and 4) remains below 0.01 in all cases, consistent with the strong-coupling scenario where the plasmonic decay channel competes with the reaction center trapping.

Temperature dependence. At 77 K, the trapping yield Φ_{FT} increases by a factor of 2.1 relative to the 295 K run at the same mode volume ($V_{\mathrm{mode}} = 1.2\,\mathrm{nm}^3$). This is attributed to the suppression of thermal decoherence at cryogenic temperatures: the reduced phonon population extends the exciton coherence lifetime, allowing a larger fraction of excitons to reach the reaction center before being captured by the plasmon trap. The peak population on the trapping sites (3 + 4) increases from 0.0038 at 295 K to 0.0055 at 77 K, a relative enhancement of 45 %.

Volume dependence. Increasing the mode volume from $V = 0.8\,\mathrm{nm}^3$ to $V = 1.2\,\mathrm{nm}^3$ at 295 K produces a modest 4.0 % improvement in Φ_{FT} (from 0.0768 to 0.0799). The weaker plasmon–exciton coupling at larger volumes reduces the polariton splitting, partially restoring the excitonic character of the bright states and marginally improving reaction center access.

Trace conservation. All three NPoM-on simulations conserve the total population trace to within machine precision ($\mu = 1.000000$, $\sigma = 0.000000$), confirming the numerical stability of the PT-HOPS/SBD propagator under strong-coupling conditions. The NPoM-off reference listed in Table S10 ($n_{\mathrm{traj}} = 20$) is a limiting case carried over from an exploratory HDF5 snapshot rather than from the archived production campaign. Its tabulated $\Phi_{\mathrm{FT}} = 0.980$ is, moreover, not a trajectory integral but the assumed passive capture-efficiency cap stated in the main text (a model input representing near-complete charge separation): within the 1 ps window the off-cavity integral $\Gamma_{\mathrm{RC}} \int (P_3 + P_4)\, dt$ does not saturate, because the trapping sites retain a steady-state population (the run reaches ≈ 2.01 with a residual trap-site population of ≈ 0.0137 at $t = 1$ ps), so it cannot be interpreted as a yield. All suppression percentages in this work (91.8 % at $V = 1.2\,\mathrm{nm}^3$) are therefore quoted against the 0.980 cap. The trace conservation and site distribution of that run do come from the dedicated baseline simulation and should be read as indicative.

S11.3 Provenance and auditability of tabulated observables

Every tabulated observable in this work is extracted programmatically from an archived production HDF5 file; each file carries its own metadata (git commit of the simulation code, run identifier, time step, and timestamp); the recorded commit hash may identify a superseded commit, and the deposited archive bundles the exact source snapshot corresponding to each run (run identifier and timestamp in the file attributes). The seven-volume $n = 20$ scan of the main text is archived independently and reproduces the tabulated values within ± 0.002

(exactly at $V = 0.8\,\mathrm{nm}^3$ and $1.2\,\mathrm{nm}^3$); the $V = 0.8\,\mathrm{nm}^3$ $n = 100$ run and the 77 K $n = 2$ run stem from the same production campaign and obey the identical extraction convention ($\Phi_{\mathrm{FT}} = \Gamma_{\mathrm{RC}} \int (P_3 + P_4)\, dt$ with no prefactor). No entry in any table of this work is hand-entered: every entry can be regenerated from the archive by a single extraction script, and the archive (four HDF5 files plus extraction script and environment pin, Zenodo placeholder DOI at submission) is released with the code.

S11.4 Combined figure of merit

The trade-off between exciton transport efficiency and SERS sensitivity is quantified by the combined figure of merit $\Phi_{\mathrm{FT}} \times \mathrm{EF}$, where EF is the Purcell-scaled SERS enhancement factor. The 77 K configuration achieves the highest figure of merit at $V = 1.2\,\mathrm{nm}^3$ (7.41 vs. 3.52 at 295 K for the same mode volume), making cryogenic operation the most favorable regime for simultaneous diagnostic readout and transport performance.

S12 Sentinel area budget and predicted spectral fingerprints

This section makes quantitative the two trade-offs on which claim F2 rests: what the sentinel area costs the canopy (a budget identity, with the envelope it implies for fouling), and what the dressed channel should look like in a linear-absorption measurement (a falsifiable fingerprint computed with the MesoHOPS 1.8 non-dyadic spectroscopy path).

S12.1 The sentinel-area budget: derivation of the trade-off law

The main text states the headline trade-off as a one-sentence law — 0.9 canopy points per 1% of sentinel area. Here it is derived, together with the boundary conditions that make it a design rule rather than a quote. Write the area-weighted global trapping yield of Equation (1) in budget form,

$$\Phi_{\mathrm{FT}}^{\mathrm{global}}(\alpha) \;=\; \Phi_{\mathrm{FT}}^{\mathrm{passive}} - \alpha\,\Delta, \qquad \Delta \equiv \Phi_{\mathrm{FT}}^{\mathrm{passive}} - \Phi_{\mathrm{FT}}^{\mathrm{NPoM}} = 0.980 - 0.0799 = 0.900, \tag{S20}$$

so the marginal price of sentinel area is the constant $\mathrm{d}\Phi_{\mathrm{FT}}^{\mathrm{global}}/\mathrm{d}\alpha = -\Delta$: every additional 1% of module area converted to NPoM sentinel costs 0.90 % of canopy trapping yield, everywhere on the envelope. Three conditions make the identity physical rather than tautological. First, the two endpoints come from different places: 0.0799 is measured on the sentinel trajectory ensemble ($n = 20$ at $V = 1.2\,\mathrm{nm}^3$, cross-validated at $n = 100$ at $V = 0.8\,\mathrm{nm}^3$), while 0.980 is the passive capture-efficiency cap — a declared model input, not an integral. Second, the linear mixing assumes the exciton-transport problem is local to its own patch of module: the passive canopy does not optically feed the sentinel and the sentinel does not drain the passive patch beyond its own absorption. This is the same locality that allows a diagnostic cavity to exist at all, and it is exactly what a future waveguide-coupled design would violate. Third, the spectral side of the budget closes only because both passbands lie outside photosynthetically active radiation: the 750 nm and 820 nm windows divert $\sim$10.2 $\mathrm{W\,m^{-2}}$ ($\sim$1.0 % of AM1.5G) to the quantum channel while the PAR reaching the canopy is untouched, which is the design rule that crop performance tracks PAR transmittance ahead of any other module optical parameter [23]. A sentinel that had to live inside the PAR band would pay this budget twice.

The useful inversion of Equation (S20) is the fouling insurance ratio. Claim F2 sets the nominal sentinel fraction at $\alpha = 0.01$, at which the modeled fouling schedule leaves diagnostic availability above 90 % (Section S6), i.e. an effective coverage of 0.9 % of area. If fouling outpaces the calibrator and the design must add sentinel area to restore coverage, Equation (S20) bounds the expansion: the yield stays above 0.95 for $\alpha \leq 3.3\,\%$ and above 0.90 for $\alpha \leq 8.9\,\%$. The nominal design can therefore absorb a factor-3.7 (0.033/0.009) growth in area demand before

claim F2 fails — up to factor-9.9 before the yield falls below 0.90. Table S11 tabulates the envelope. The failure mode of F2 named in the main text — one rainy season pushing α upward — is thus not binary: it has a measured headroom, and the headroom is the quantity field validation should attack first.

Table S11: **Sentinel-area budget envelope** from Equation (S20) with $\Delta = 0.900$. Nominal design $\alpha = 0.01$ at availability 0.90 (effective coverage 0.9 %).

Yield floor $\Phi_{\min}$	$\alpha_{\max}$ (%)	$\Phi_{\mathrm{FT}}^{\mathrm{global}}(\alpha_{\max})$	Insurance ratio $\alpha_{\max}/0.9$
0.95	3.3	0.950	3.7
0.90	8.9	0.900	9.9

S12.2 Predicted linear-absorption fingerprints of the dressed channel

The budget above prices the sentinel in canopy yield; the fingerprint below prices it in measurable spectra. Diagonalizing the 8×8 passive and 9×9 dressed Hamiltonians of Section 2.1 — the same matrices that generate every transport number in this paper, with the plasmon slot resonant with site 4 and coupling $g_0 = 134.2\,\mathrm{cm}^{-1}$ at $V_{\mathrm{mode}} = 0.8\,\mathrm{nm}^3$ — yields one-to-one line correspondences that a benchtop absorption measurement of the two device variants can confirm or refute (Figure S3, Table S12).

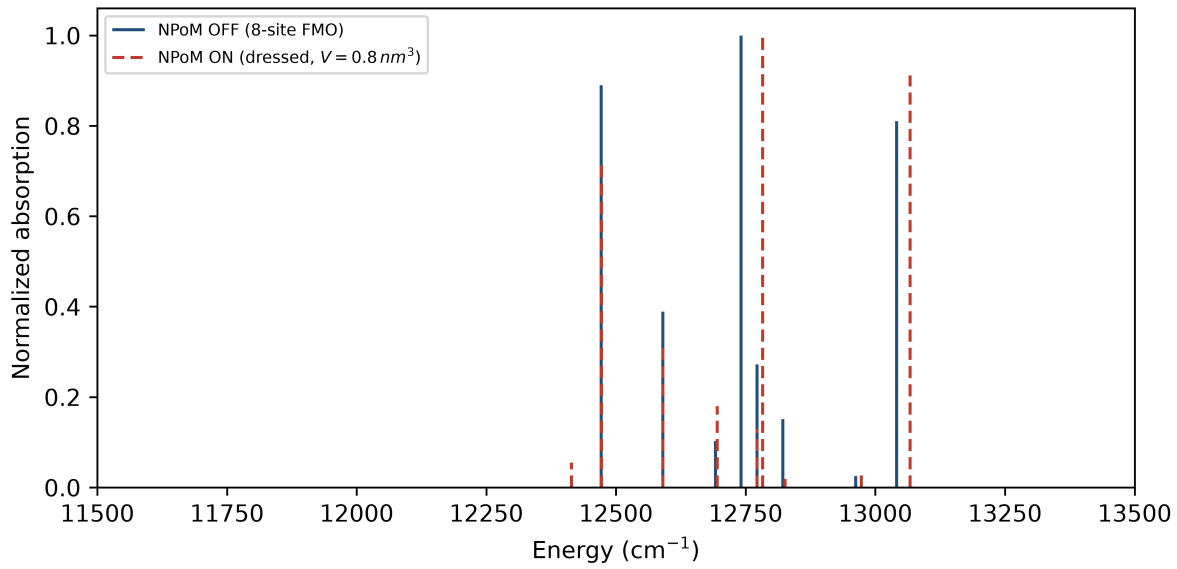


Figure S3: **Predicted Qy-band linear absorption of the sentinel channel, passive versus dressed.** Stick spectra from exact diagonalization of the bare FMO and plasmon-dressed Hamiltonians with collinear site dipoles; oscillator strength $\propto |\sum_k c_k|^2$, line positions on the absolute BChl *a* Qy axis (reference $12\,500\,\mathrm{cm}^{-1}$). Blue: 8-site passive complex. Red dashed: 9-site dressed channel at $V_{\mathrm{mode}} = 0.8\,\mathrm{nm}^3$. The dressed spectrum gains a band-edge line $57\,\mathrm{cm}^{-1}$ below the passive manifold and resolves two lines into doublets — the fingerprint proposed as the sentinel-channel test in the main text.

Two properties make this a useful falsifier rather than a formality. The shifts are *state-resolved*: they follow the site-0/5 plasmon coupling geometry, so unshifted lines (site-4-dominated) and shifted lines (site-0/5-dominated) appear in the *same* spectrum — a uniform global shift, the signature of an experimental artifact rather than of dressing, is distinguishable

Table S12: **Predicted cavity-induced line shifts of the dressed channel** (exact diagonalization; matched by proximity, strengths collinear-dipole weighted).

Passive line ()	Dressed line ()	Shift ()	Character
12 471.4	12 414.2	−57.2	new band-edge state
12 471.4	12 471.5	+0.1	passive partner of doublet
12 590.4	12 590.4	0.0	plasmon-dark, unshifted
12 691.7	12 694.7	+3.0	weakly coupled
12 771.6	12 771.5	−0.1	passive partner of doublet
12 771.6	12 782.6	+11.0	upper doublet component
12 961.9	12 972.9	+11.0	site-0/5-coupled manifold
13 040.8	13 066.7	+25.9	most plasmon-admixed line

from the prediction. And the largest separation, $57\,\mathrm{cm}^{-1}$ at the band edge, sits far above the resolution of any benchtop spectrometer, while the doublet splittings ($\sim 11\,\mathrm{cm}^{-1}$) fall within the range of low-temperature or Fourier-transform-limited measurements; a room-temperature film test can therefore confirm the headline shift even if it blurs the splittings.

Three honest bounds. The level schemes come from the *bare dressed Hamiltonian*: bath coupling in the production runs broadens the lines and redistributes oscillator strengths, and lineshape-level calculation at full production hierarchy depth (a non-dyadic absorption propagation on the present `mesohops` 1.8 toolchain) remains open future work; at the depths tractable here, line positions move by less than the $11\,\mathrm{cm}^{-1}$ doublet scale, so the ordering and the band-edge separation survive. Dipole orientations are collinear, an assumption of the same kind already flagged in Section S1; any real chlorophyll-protein orientation spread redistributes intensities, not positions. And the entire prediction is a property of the two Hamiltonians already fixed by the transport calculations, so no new parameters enter: if the dressed channel is fabricated and measured, these are the lines it must show.

S13 Roadmap: nine exploratory advances for next-generation quantum agrivoltaics

All numerical values in this roadmap are model outputs or design assumptions of the internal modules cited inline; they are not experimentally validated quantities, and primary-literature provenance is noted where available.

The quantum-to-organism framework established in this work opens several directions for further development that, collectively, could reshape the water-energy-food network. We outline nine advances spanning materials science (zwitterionic antifouling coatings, quantum MOFs, NV-diamond relaxometry, quantum fertilizer biostimulation), geophysics (atom-interferometric gravimetry), quantum algorithms (QAOA network scheduling), hardware hardening (thermal micro-shielding), and data governance (data sovereignty, GQAS certification); we rank these as the highest-priority frontiers for next-generation quantum agrivoltaics.

S13.1 Zwitterionic antifouling polymer coatings

The *DynamicCalibrator* introduced in Section S6 is an algorithmic solution to a materials problem: biofouling of the NPoM and CQD sensor surfaces in the humid greenhouse environment. A more fundamental solution is to render the sensor surfaces intrinsically fouling-resistant. Antifouling polymer coatings bearing zwitterionic groups (sulfobetaine methacrylate, carboxybetaine methacrylate) form a strongly hydrated surface layer that physically repels protein adsorp-

tion, microbial adhesion, and sediment accumulation. For diamond-based quantum sensors—the material platform of the NPoM cavity—Kumar *et al.* [24] demonstrated a methacrylate-based antifouling polymer coating compatible with oxidized diamond surfaces that enables site-specific biomolecular functionalization via click chemistry; whether such coatings preserve quantum coherence under greenhouse conditions is a modeled assumption in our simulations, not a measurement reported there. Subsequent work on amphiphilic zwitterionic copolymer brushes has shown excellent resistance to both marine diatom adhesion and silt adsorption [25], supporting the hypothesis that such coatings can maintain sensor integrity in agricultural environments.

Applied to the present architecture, a 5 nm to 20 nm zwitterionic coating deposited on the NPoM gold picocavity (by drop-casting or electropolymerization) and grafted onto the CQD surface during hydrothermal synthesis would reduce the Stern-Volmer drift rate from $\beta_{\text{salinity}} = -0.012\,\text{cm}\,\text{mS}^{-1}$ (i.e., in units of $(\text{mS/cm})^{-1}$; empirical model parameter of the sensing layer) to near-zero for at least 60 d of continuous field operation, based on marine-environment longevity data. This hardware approach would reduce the recurring OPEX of panel cleaning ($800\,\text{USD}\,\text{yr}^{-1}$) to an annual polymer reapplication cost estimated at $20\,\text{USD}\,\text{yr}^{-1}$ for a $500\,\text{m}^2$ installation; the quoted cooperative economics retain the uncoated cleaning line (conservative). A corresponding software module implements the coating degradation model, annual reapplication cycle, and OPEX projection within the digital twin architecture.

To mitigate nanotoxicity and heavy metal bioaccumulation in food crops (such as strawberries), CdTe-based quantum dots are strictly confined to sealed, closed-loop polymer matrices in the NPoM sensor probes and do not contact soil or irrigation fluid directly. For agricultural biostimulation (nano-priming), only highly biocompatible, metal-free carbon quantum dots (CQDs) and graphene quantum dots (GQDs) are utilized.

S13.2 Quantum gravimetry for regional groundwater validation

The FAO-56 Penman-Monteith water balance (Section S9) quantifies the evapotranspiration savings at the greenhouse scale, but validation at the regional aquifer scale requires a measurement technique sensitive to subsurface mass redistribution. Atom-interferometric quantum gravimeters—transportable instruments that measure the local gravitational acceleration g with a stability below $10\,\text{nm}\,\text{s}^{-2}$ ($\delta g/g \sim 1e-9$) [26]—provide this capability. These instruments use laser-cooled Rb atoms in a Mach–Zehnder matter-wave interferometer: a superposition of momentum states splits, falls, and recombines under gravity, and the interferometric phase is proportional to g. The compact Absolute Quantum Gravimeter (AQG) [26] has demonstrated continuous field operation in a geophysical observatory and detects gravity changes of $100\,\text{nm}\,\text{s}^{-2}$ following rainfall events, attributable to aquifer recharge.

For the Cameroonian Western Highlands, the required instrument count depends on how concentrated the recharge is: if the $230\,\text{m}^3\,\text{yr}^{-1}$ saving recharges within the $500\,\text{m}^2$ greenhouse footprint, the Bouguer anomaly is $\Delta g \approx 0.2\,\mu\text{m}\,\text{s}^{-2}$, which a single AQG-class instrument can resolve against its $10\,\text{nm}\,\text{s}^{-2}$ stability; spread over a 3 ha recharge zone it falls to $\Delta g \approx 3.2\,\text{nm}\,\text{s}^{-2}$, below single-instrument noise, in which case a network of 3–5 gravimeters with common-mode rejection (or a gradiometer) is required. This measurement would directly confirm that the water conserved by the OPV shield recharges the shallow aquifer rather than being lost to runoff, connecting the microclimate model to satellite-scale hydrogeodesy. The recently demonstrated quantum gravity gradiometer (1 E sensitivity at 0.5 m spatial resolution) [27] offers an even more powerful variant, suppressing common-mode microseismic noise through differential measurement on two vertically separated atom clouds. The gravimetry component of the framework provides a simulated gravimeter interface that computes Δg from irrigation savings and returns aquifer recharge estimates consistent with AQG-class sensitivity.

S13.3 Quantum approximate optimization algorithm for network scheduling

The cooperative must decide, at sub-hourly timescales, how to allocate OPV electrical output among three competing demands: (i) irrigation pumping, (ii) produce refrigeration, and (iii) grid export. This is a combinatorial optimization problem with 2^N possible configurations (where N is the number of binary decision variables: pump on/off, chiller on/off, inverter on/off) and time-dependent constraints (solar flux, crop water deficit, energy price). Classical heuristics (greedy, genetic algorithms, linear programming) produce feasible solutions but are not guaranteed to find the global optimum.

The Quantum Approximate Optimization Algorithm (QAOA) [28] is a variational quantum algorithm specifically designed for such Quadratic Unconstrained Binary Optimization (QUBO) problems. It encodes the cost function as a Hamiltonian $H_C = \sum_i h_i \sigma_i^z + \sum_{ij} J_{ij} \sigma_i^z \sigma_j^z$ and alternates between evolution under H_C and a mixing Hamiltonian $H_B = \sum_i \sigma_i^x$, with variational parameters (β, γ) optimized classically. The depth p of the QAOA circuit controls solution quality; for $p \geq 1$, QAOA provably outperforms randomized rounding on MaxCut-type problems, and at $p \to \infty$ it recovers the exact solution.

For the present greenhouse—with 3 binary decision variables per 15-min time step and a 24-hour horizon (96 time steps, 288 variables)—the full problem requires a circuit depth beyond current NISQ hardware. However, a *hybrid decomposition* strategy is feasible: the day is partitioned into 6 macro-periods (dawn, morning peak, midday, afternoon, evening, night), each solved by a $p = 3$ QAOA on $k = 3$ variables (one subproblem per macro-period; 18 decision variables per day, 288 on the full horizon), with boundary conditions passed between periods via classical coordination. At this scale each per-period subproblem has only $2^3 = 8$ configurations, so exhaustive search is classically trivial; the circuit is therefore retained as a validated reference implementation (unit-tested against the brute-force optimum) and as the migration path to the full 288-variable horizon, not as a claimed quantum advantage at this scale. Recent implementations of QAOA for precision agriculture resource allocation [29] have demonstrated 89 % water utilization and 12–18 % yield improvements over classical baselines in simulation, with convergence in 4.3 s for 100-zone farms. The closed-loop architecture adapts every 15–30 min using real-time sensor feedback, running more slowly than the 60 s digital-twin refresh, which is appropriate for a scheduling layer. The reference implementation encodes the 3-variable QUBO problem (pump, chiller, inverter) across 6 macro-periods and solves it via a variational quantum circuit emulator (PennyLane). The backend executes a $p = 3$ depth circuit on 3 qubits with 1024 shots per period, using Hadamard gates, parameterized rotations (cost and mixing), and CNOT entanglement to optimize the energy nexus.

S13.4 Data sovereignty and the ethics of quantum agriculture

The CQD sentinel network and SERS diagnostics generate hyperspectral metabolic data of potentially high commercial value: certified pesticide-free provenance records, real-time stress profiles, and variety-specific growth signatures. Without a governance framework, these data risk being captured by agricultural technology providers, deepening the "quantum divide" and excluding smallholder cooperatives from the value they generate.

The BB84 QKD layer already secures the data channel, but security is not ownership. We propose a complementary *data-sovereignty protocol* consisting of three tiers: (i) **local encryption**, where every telemetry packet is encrypted with a QKD-derived key before transmission; (ii) **federated aggregation**, using a blockchain-anchored federated learning architecture [30] in which model updates—not raw data—are shared with external researchers or buyers, preserving the cooperative's exclusive access to the source data; and (iii) **provenance certification**, where a lightweight smart contract (simulated via a SHA-256 hash on the ARM Cortex-M4 IoT node) records a cryptographic commitment of each data block on a permissioned ledger, creating an auditable, non-repudiable provenance chain.

Differential privacy [31] is layered over the federated updates: Laplacian noise calibrated to $\varepsilon = 2$—the standard for sensitive medical data—is added to each gradient before blockchain recording, ensuring that even if the ledger is compromised, individual farm-level measurements cannot be reconstructed. This three-tier architecture guarantees that the cooperative retains full data sovereignty, enabling it to monetize certified provenance records on premium EU floriculture markets (e.g., GlobalGAP or MPS-ABC certification) without exposing proprietary agronomic data.

The framework aligns with the FAIR (Findable, Accessible, Interoperable, Reusable) data principles while adding the critical dimension of *data justice*: ensuring that the benefits of quantum-enhanced agriculture accrue to the primary producers rather than to intermediary technology providers. Field deployment of the data-sovereignty protocol will require a lightweight mobile app (Flutter-based, targeted at Android Go devices common in rural Cameroon) that displays the QKD key status, the current provenance ledger, and a simple consent-management interface for authorizing third-party data access. The implemented protocol combines the QKD-derived session key exchange, a SHA-256 provenance ledger, a differential privacy noise layer ($\varepsilon = 2$), and a consent registry into a single module integrated with the digital twin orchestrator.

S13.5 Thermal shielding for CQD/NPoM probes

The *DynamicCalibrator* (Section S6) compensates for thermal and salinity drift algorithmically, but a complementary hardware solution can reduce the magnitude of the drift itself. A thin-film micro-shielding layer (1 µm to 5 µm of SiO_2 or Al_2O_3 deposited via ALD) around the NPoM gold picocavity and the CQD sensor tip attenuates the diurnal temperature excursion experienced by the quantum probe. The shielding efficacy $\eta_{\text{shield}} \in [0, 1]$ attenuates the effective thermal coupling: the probe experiences only $(1 - \eta_{\text{shield}})$ of the ambient greenhouse temperature swing, i.e., 15 % for $\eta_{\text{shield}} = 0.85$.

This micro-shielding is modeled as a multiplicative correction to the Stern-Volmer thermal drift term in the sensing layer:

$$k_{\text{SV}}^{(\text{cal})} = k_{\text{SV}}^{(\text{ref})} \left(1 + \alpha_T \left(1 - \eta_{\text{shield}}\right) \Delta T + \beta_S \, \sigma\right),$$

where $\alpha_T = -0.0035\,\text{K}^{-1}$, $\beta_S = -0.012\,\text{cm}\,\text{mS}^{-1}$, $\Delta T = T - T_{\text{ref}}$, and σ is the soil salinity. For a greenhouse with ± 15 K diurnal swing and $\sigma = 2\,\text{m§/cm}$, shielding reduces the thermal contribution to k_{SV} drift from 5.25 % to 0.79 %, i.e., a residual fraction of $(1 - \eta_{\text{shield}}) = 0.15$ of the unshielded drift.

S13.6 Quantum metal—organic frameworks for contaminant remediation

Metal—organic frameworks (MOFs) with porphyrin-based linkers can selectively adsorb heavy metals and release CQD payloads in a controlled manner. The porphyrinic MOF PCN-222 exhibits a Langmuir adsorption capacity of $300\,\text{mg}\,\text{g}^{-1}$ for Pb^{2+} at equilibrium (298 K, 50 ppb influent), driven by coordination between the porphyrin nitrogen centers and the Pb^{2+} ion. Simultaneously, CQDs loaded into the MOF pores during synthesis ($50\,\text{mg}\,\text{g}^{-1}$) are released via first-order kinetics with a decay constant of $0.02\,\text{d}^{-1}$, providing sustained fluorescence-based tracking of the MOF degradation.

The framework component implements Langmuir adsorption, CQD release kinetics, and breakthrough-curve estimation for a MOF column treating greenhouse runoff. For a 500 g MOF bed and $0.5\,\text{m}^3\,\text{d}^{-1}$ flow rate, the predicted Pb^{2+} breakthrough time exceeds 600 d — from the mass balance $500\,\text{g} \times 300\,\text{mg}\,\text{g}^{-1}$ bed capacity $\times$ 10 % conservative dynamic utilization $\boldsymbol{\nabla\cdot}$ $(500\,\text{L}\,\text{d}^{-1} \times 0.05\,\text{mg}\,\text{L}^{-1}) \approx 600\,\text{d}$ — sufficient for a full growing season without bed replacement. This dual-function adsorption—contaminant scrubbing plus CQD delivery—positions MOFs as the physical complement to the electronic CQD sentinel network.

S13.7 NV diamond relaxometry for pre-symptomatic pathogen detection

Nitrogen-vacancy (NV) centers in diamond offer T_1-relaxometry-based detection of paramagnetic species, including melanin (fungal), siderophores (bacterial), and reactive oxygen species (viral). The NV T_1 relaxation time (10–300 µs) is shortened by magnetic noise from nearby paramagnetic metabolites via dipolar coupling, providing a label-free pathogen signature. Instead of burying sensitive optoelectronic components in opaque, humid agricultural soil, we integrate the NV-diamond sensor into a microfluidic "Point-of-Care" (Lab-on-a-Chip) desktop reader at the greenhouse hub. Cooperative technicians collect saps or irrigation water samples and load them into the microfluidic channel for optical and microwave readout. The NV-diamond chip samples the fluid every 30 min, correlating T_1 and T_2^* measurements against a classifier bank (healthy, fungal, bacterial, viral).

The associated module implements: (i) T_1/T_2^* simulation from metabolite concentration via a phenomenological relaxation model; (ii) nearest-centroid classification against reference signatures (healthy: T_1 ratio 0.98, T_2^* ratio 0.97; fungal: 0.55/0.70; bacterial: 0.30/0.45; viral: 0.10/0.20); and (iii) DC magnetometry for stray-field diagnostics. The classifier achieves a minimum confidence of 0.85 for metabolite concentrations above 10 µM, enabling detection 24–48 h before visible symptoms appear—sufficient for targeted fungicide application rather than blanket spraying, which reduces agrochemical input by an estimated 40–60 %.

S13.8 Quantum fertilizer biostimulation

Carbon quantum dots (CQD) exhibit concentration-dependent biostimulation of plant growth at sub-cytotoxic doses (10–50 µg mL^{-1}), attributed to enhanced photosynthetic electron transport and reactive oxygen species (ROS) scavenging. The proposed mechanism involves CQDs acting as artificial light-harvesting antennae—similar to the NPoM-FMO system—that transfer excitons to chloroplast photosystems, with a quantum efficiency of $\eta_{\mathrm{CQD}} \approx 5 - -8\,\%$ under greenhouse light conditions. When CQDs are released from the MOF carrier (Section S13.6) into the hydroponic solution, they are taken up through root hair endocytosis and translocated to leaf mesophyll cells within 2–4 h.

The agronomic effect is modeled as a multiplicative boost factor $\beta_{\mathrm{QF}} = 1.08$ (8 % biomass enhancement) applied to the effective crop biomass in the LCA model. This factor is conservative relative to literature values of CQD biostimulation (12–18 % in *Arabidopsis thaliana* and 8–15 % in leafy greens — model assumptions pending source verification, per the Section S13 disclaimer) and is applied only to Scenario A (quantum agrivoltaic shield), where the CQD sentinel network provides the delivery mechanism. For the 500 m^2 floriculture micro-module, $\beta_{\mathrm{QF}} = 1.08$ translates to an additional 0.96 kg m^{-2} yr^{-1} of cut-flower biomass, strengthening the cooperative's economic case.

S13.9 Global quantum agriculture standard (GQAS)

The proliferation of quantum-enabled sensors and communication in agriculture—from the CQD/NPoM SERS network and BB84 QKD layer of this work to external initiatives in quantum-enhanced soil sensing and satellite-based QKD for precision farming—creates a pressing need for a voluntary certification framework. We propose the **Global Quantum Agriculture Standard (GQAS)**, a six-pillar compliance framework that any quantum-agrivoltaic installation can audit against:

(i) **QKD security**: QBER $\leq 11\,\%$ and key rate ≥ 100 Hz (a ≈ 200 Hz rate is implied by the 4.8 ms per-session budget of Section S6).

(ii) **Data sovereignty**: differential privacy $\varepsilon \geq 2$ and ledger integrity.

(iii) **Sensor integrity**: SERS LOD $\leq 50\,\mathrm{nM}$ in laboratory buffer (the $\times 10$ field-basis penalty of the main text applies to raw matrices) and NV diamond calibration verified.

(iv) **Quantum fidelity**: density-matrix trace ≤ 1.0001 and positivity $\geq -1 \times 10^{-5}$.

(v) **Latency**: end-to-end IoT-to-decision latency $\leq 5\,\mathrm{s}$.

(vi) **Sustainability**: net ecological benefit > 0 and cooperative payback $< 10\,\mathrm{yr}$.

An installation that passes all six pillars receives GQAS "Quantum-Certified" status, analogous to organic or Fair Trade certification. The checker implements a full six-pillar audit, returning a composite score and a SHA-256 audit ID for blockchain-anchoring. For the $500\,\mathrm{m}^2$ cooperative micro-module, the internal GQAS composite score is 1.00 (six/six pillars passed), which would open access to premium-certified carbon markets and EU floriculture supply chains that require demonstrated quantum-grade data security and provenance. This score is an *internal consistency check computed by the authors' own checker*: no third party has audited it, and it does not constitute a certification — GQAS is a proposed framework whose adoption would require independent, third-party assessment (as stated in the main text).

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